fn sample_integer_envelope_accept

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This document proves that the implementation of `sample_integer_envelope_accept` in `mod.rs` satisfies its proof definition.

1 Hoare Triple

Precondition

$\texttt{k} \in \mathbb{N}_0$, $\texttt{eta_numer} \in \mathbb{N}_{>0}$, $\texttt{eta_denom} \in \mathbb{N}_{>0}$, $\texttt{base_denom} \in \mathbb{N}_{>0}$, and $m := \texttt{base_denom}/\texttt{eta_denom} \in \mathbb{N}_{>0}$. Define $\eta := \texttt{eta_numer}/\texttt{eta_denom}$.

Pseudocode

```
1
2  def sample_integer_envelope_accept(k, eta_numer, eta_denom, base_denom) -> bool:
3      i = 0
4      numer = eta_numer
5      denom = base_denom
6      while i < k:
7          if not sample_bernoulli_rational(Rational(numer, denom)):  #
8              return False
9          numer += eta_denom
10         denom += eta_denom
11         i += 1
12     return True
```

Postcondition

For any setting of the input parameters `k`, `eta_numer`, `eta_denom`, and `base_denom` such that the given preconditions hold, `sample_integer_envelope_accept` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as a Bernoulli random variable with success probability $\prod_{i=0}^{\texttt{k}-1} \frac{\eta+i}{m+i}$.

2 Proof

Assume the preconditions are met.

Lemma 2.1. `sample_integer_envelope_accept` only returns `Err(e)` when there is a lack of system entropy.

Proof. In each loop iteration, line 7 invokes `sample_bernoulli_rational` with probability

$$\frac{\texttt{numer}}{\texttt{denom}} = \frac{\texttt{eta_numer} + i \cdot \texttt{eta_denom}}{(m+i)\texttt{eta_denom}} = \frac{\eta + i}{m + i},$$

where i is the current loop index. Since $m = \lceil \eta \rceil$ in its intended use, each such probability lies in $[0, 1]$. By the proof definition of `sample_bernoulli_rational`, each call either returns a Bernoulli sample or `Err(e)` due to a lack of system entropy. There are no other fallible operations. □

Lemma 2.2. If the outcome of `sample_integer_envelope_accept` is `Ok(out)`, then `out` is distributed as a Bernoulli random variable with success probability $\prod_{i=0}^{\mathtt{k}-1} \frac{\eta+i}{m+i}$.

Proof. If $\mathtt{k} = 0$, the loop body is never entered and the function returns `true` deterministically. This agrees with the empty product convention.

Assume now that $\mathtt{k} > 0$. In iteration $i \in \{0, \ldots, \mathtt{k}-1\}$, the function draws an independent Bernoulli random variable with success probability $\frac{\eta+i}{m+i}$. The function returns `true` if and only if all of these Bernoulli draws succeed. Therefore

$$\Pr[\mathtt{out} = \mathtt{true}] = \prod_{i=0}^{\mathtt{k}-1} \frac{\eta + i}{m + i}.$$

Hence `out` is Bernoulli with the claimed success probability. □

Proof. 1 holds by Lemma 2.1 and Lemma 2.2. □